Chapter 8 Advanced Counting Techniques

8.1 Recurrence Relations

8.2 Solving Linear Recurrence Relations

8.4 Generating Functions

8.5 Inclusion-Exclusion

8.6 Applications of Inclusion-Exclusion



Generating functions are useful for manipulating sequences.

- **It can be used to solve many kinds of counting problems, such as**

  -- The problem of combination and permutation with constraints

  -- The number of ways to make change for a dollar using coins of different denominations

- **Solve recurrence relations**

- **Prove the combinatorial identities**



1. Generating function for a sequence

【Definition 1】 The *generating function* for the sequence $a_0, a_1, a_2, \cdots, a_k, \cdots$ of real numbers is the infinite series

$$G(x) = a_0 + a_1 x + \ldots + a_k x^k + \ldots = \sum_{k=0}^{\infty} a_k x^k$$

A generating function is a clothesline on which we hang up a sequence of numbers for display.

— Herbert Wilf, *Generatingfunctionology* (1994)



〖Example 1〗

(1) What is the generating function for the sequence 1, 1, 1, 1, ...?

$$G(x) = 1 + x + x^2 + x^3 + \cdots = \sum_{k=0}^{\infty} x^k = \frac{1}{1-x} \quad for |x| < 1$$

(2) What is the generating function for the sequence 0, 1, 2, 3, 4, 5, ...?

$$G(x) = \sum_{k=0}^{\infty} kx^k$$



The generating function for a finite sequence of real numbers

$a_0, a_1, a_2, \cdots, a_n$ **is**

$$G(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$

〖Example 2〗

(1) The finite sequence: 1,1,1. The generating function for this sequence is

$$G(x) = 1 + x + x^2 = \frac{1 - x^3}{1 - x}$$

(2) Let $a_k = C(m,k), k = 0,1,2,\cdots,m$**. The generating function for this sequence is**

$$G(x) = C(m,0) + C(m,1)x + C(m,2)x^2 + \cdots + C(m,m)x^m = (1+x)^m$$



2. Useful Facts About Power Series

【Theorem 1】Let $f(x)=\sum_{k=0}^{\infty}$

(1) $f(x)+g(x)=\sum_{k=0}^{\infty}(a_k+b_k)x^k$

(2) $\alpha\cdot f(x)=\sum_{k=0}^{\infty}\alpha\cdot a_k x^k$ α

(3) $x\cdot f'(x)=\sum_{k=0}^{\infty}k\cdot a_k x^k$

(4) $f(\alpha x)=\sum_{k=0}^{\infty}\alpha^k\cdot a_k x^k$

(5) $f(x)g(x)=\sum_{k=0}^{\infty}(\sum_{j=0}^{k}a_j b_{k-j})x^k$

Proof:

$$\sum_{k=0}^{\infty}ka_k x^k = \sum_{k=0}^{\infty}a_k\cdot x\cdot kx^{k-1}$$

$$=x\sum_{k=0}^{\infty}a_k(x^k)'$$

$$=x(\sum_{k=0}^{\infty}a_k x^k)'$$

$$=xf'(x)$$



2. Useful Facts About Power Series

Let $f(x) = \sum_{k=0}^{\infty} a_k x^k$

(1) $f(x) + g(x) =$

(2) $\alpha \cdot f(x) = \sum_{k=0}^{\infty}$

(3) $x \cdot f'(x) =$

(4) $f(\alpha x) = \sum_{k=0}^{\infty}$

(5) $f(x)g(x) = \sum$

Proof:

$$\sum_{k=0}^{\infty}\left(\sum_{j=0}^{k} a_j b_{k-j}\right)x^k$$

$$= a_0b_0 + (a_0b_1 + a_1b_0)x + (a_0b_2 + a_1b_1 + a_2b_0)x^2$$

$$+ \ldots + \left(\sum_{j=1}^{k} a_j b_{k-1}\right)x^k + \ldots$$

$$= (a_0 + a_1x + a_2x^2 + \ldots)(b_0 + b_1x + b_2x^2 + \ldots)$$

$$= f(x) \cdot g(x)$$



Using the above properties, the generating functions of some sequences can be obtained easily.

〖Example 3〗 (1) What is the generating function for the sequence 0,1,2,3,4,5,…?

Solution:

$$b_k = k$$

$$G(x) = \sum_{k=0}^{\infty} kx^k$$

$$= x\left(\frac{1}{1-x}\right)'$$

$$= \frac{x}{(1-x)^2}$$



〖Example 3〗 (2) Suppose that the generating function of the sequence: $a_0, a_1, a_2, \cdots, a_n, \cdots$ is $G(x)$. What is the generating function for the sequence $b_k = \sum_{i=0}^{k} a_i$?

Solution:

$a_k \leftrightarrow G(x)$,

$c_k = 1$

$b_k = \sum_{i=0}^{k} a_i$

$= \sum_{i=0}^{k} a_i \times c_{k-i}$

$F(x) = G(x) \cdot \frac{1}{1-x}$

For example:

$1,1,1,\cdots \longleftrightarrow \frac{1}{1-x}$

$1,2,3,4,\cdots,k+1,\cdots \longleftrightarrow \frac{1}{(1-x)^2}$

〖Example 3〗 (3) What is the generating function for the sequence $a_k = k^2$?

Solution:

$$a_k = 1 \quad \longleftrightarrow \quad \frac{1}{1-x}$$

$$a_k = k \quad \longleftrightarrow \quad \frac{x}{(1-x)^2}$$

$$a_k = k^2 \quad \longleftrightarrow \quad x\left(\frac{x}{(1-x)^2}\right)' = \frac{x(1+x)}{(1-x)^3}$$



〖Example 3〗(4) What is the generating function for the sequence $a_k = \sum_{i=1}^{k} i^2$?

Solution:

$$a_k = k^2 \quad \longleftrightarrow \quad x\left(\frac{x}{(1-x)^2}\right)' = \frac{x(1+x)}{(1-x)^3}$$

$$a_k = \sum_{i=1}^{k} i^2 \quad \longleftrightarrow \quad \frac{x(1+x)}{(1-x)^4}$$



〖Example 4〗 Let $f(x)=\dfrac{1}{1-4x^2}$ **. Find the coefficient** $a_0,a_1,a_2,\cdots,a_n,\cdots$ **in the expansion** $f(x)=\sum_{k=0}^{\infty}a_k x^k$ **.**

Solution:

$$f(x)=\frac{1}{1-4x^2}=\frac{1}{(1-2x)(1+2x)}=\frac{1}{2}\left(\frac{1}{1-2x}+\frac{1}{1+2x}\right)$$

$$\frac{1}{1-2x}\leftrightarrow 2^k \qquad \frac{1}{1+2x}\leftrightarrow (-2)^k$$

$$a_k=\frac{1}{2}(2^k+(-2)^k)=\begin{cases}2^k & k \text{ is even}\\ 0 & k \text{ is odd}\end{cases}$$



❋ The extended binomial coefficient

Recall:

$$\binom{m}{k} = C(m,k) = \frac{m!}{k!(m-k)!}$$

Where *m*, *k* are nonnegative integers, *k*≤*m*

【Definition 2】 Let *u* be a real number and *k* a nonnegative integer. Then the *extended binomial coefficient* is defined by

$$\binom{u}{k} = \begin{cases} u(u-1)\cdots(u-k+1)/k! & \text{if} \quad k>0 \\ 1 & \text{if} \quad k=0 \end{cases}$$



〖**Example 5**〗 $(1)\binom{1/2}{3}=?$ $(2)\binom{-n}{k}=?$

Solution:

$$(1)\binom{1/2}{3}=\frac{(1/2)(1/2-1)(1/2-2)}{3!}=1/16$$

$$(2)\binom{-n}{k}=\frac{(-n)(-n-1)...(-n-k+1)}{k!}$$

$$=\frac{(-1)^k n(n+1)...(n+k-1)}{k!}$$

$$=(-1)^k C(n+k-1,k)$$



The extended Binomial Theorem

【 Theorem 2】Let *x* be a real number with $|x|<1$ and let *u* be a real number. Then

$$(1+x)^u = \sum_{k=0}^{\infty} \binom{u}{k} x^k$$



〖Example 6〗Find the generating functions for

$$(1+x)^{-n} \; and \; (1-x)^{-n}$$

where *n* is a positive integer, using the extended Binomial Theorem.

Solution:

By the extended Binomial Theorem , it follows that

$$(1+x)^{-n}$$

$$= \sum_{k=0}^{\infty} \binom{-n}{k} x^k$$

$$= \sum_{k=0}^{\infty} (-1)^k C(n+k-1,k) x^k$$

$$(1-x)^{-n}$$

$$= \sum_{k=0}^{\infty} \binom{-n}{k} (-x)^k$$

$$= \sum_{k=0}^{\infty} (-1)^k C(n+k-1,k)(-1)^k x^k$$

$$= \sum_{k=0}^{\infty} C(n+k-1,k) x^k$$



3. Some Common Used Generating Functions

Sequence	Generating function
(1) $C(n,k)$	$\sum_{k=0}^{n} C(n,k)x^k = (1+x)^n$
(2) $C(n,k)a^k$	$(1+ax)^n$
(3) $\overbrace{1,1,\cdots,1}^{n}$	$1+x+x^2+\cdots+x^n = \frac{1-x^{n+1}}{1-x}$
(4) $1,1,1,\cdots$	$\frac{1}{1-x}$
(5) a^k	$\frac{1}{1-ax}$
(6) $k+1$	$\frac{1}{(1-x)^2}$



3. Some Common Used Generating Functions

Sequence	Generating function
(7) $C(n+k-1,k)$	$(1-x)^{-n}$
(8) $(-1)^k C(n+k-1,k)$	$(1+x)^{-n}$
(9) $C(n+k-1,k)a^k$	$(1-ax)^{-n}$
(10) $\frac{1}{k!}$	e^x
(11) $\frac{(-1)^{k+1}}{k}$	$\ln(1+x)$



4. Counting with Generating Functions

Generating functions can be used to solve a wide variety of counting problems, such as

- ✓ **Count the number of combinations from a set when repetition is allowed and additional constraints exist.**
- ✓ **Count the number of permutations**



〖Example 7〗Use generating functions to find the number of *r*-combinations from a set with *n* elements with repetitions allowed.

Solution:

Since there are *n* elements in the set, each can be selected zero times, one times and so on. It follows that

$$G(x)=(1+x+x^2+x^3+\cdots)^n=(\frac{1}{1-x})^n=\frac{1}{(1-x)^n}$$

the number of *r*-combinations from a set with *n* elements with repetitions allowed, is the coefficient a_r of x^r in the expansion of *G*(*x*) . Since

$$\frac{1}{(1-x)^n}=\sum_{k=0}^{\infty}C(n+k-1,k)x^k$$

Then the coefficient a_r equals *C* (*n*+*r*-1,*r*)



〖Example 8〗Find the number of solutions of $e_1 + e_2 + e_3 = 17$ **where** e_1, e_2, e_3 **are nonnegative integers with** $2 \le e_1 \le 5, 3 \le e_2 \le 6, and\ 4 \le e_3 \le 7$ **.**

Solution:

$$e_1 + e_2 + e_3 = 17$$

The generating function for this counting problem is

$$G(x) = (x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7)$$

The number of solutions is the coefficient of x^{17} in the expansion of $G(x)$.



〖Example 9〗Suppose that there are 2*r* red balls, 2*r* blue balls, and 2*r* white balls. How many ways to select 3*r* balls from these balls?

Solution:

$$G(x)=(1+x+x^2+\cdots+x^{2r})^3$$

The coefficient a_{3r} of x^{3r} in the expansion o

How to find a_{3r} ?

of this problem.

$$G(x)=(1+x+x^2+\cdots+x^{2r})^3=\left(\frac{1-x^{2r+1}}{1-x}\right)^3=\frac{1-3x^{2r+1}+3x^{4r+2}-x^{6r+3}}{(1-x)^3}$$

$$\because F(x)=\frac{1}{(1-x)^3}$$

The coefficient of term x^i in $F(x)$ is $C_{3+i-1}^{i}=C_{i+2}^{i}$

$\because 2r+1+i=3r \qquad \therefore i=r-1$

The coefficient of term x^{r-1} in $F(x)$ is C_{r+1}^{r-1}

$$\therefore a_{3r}=C_{3r+2}^{3r}-3C_{r+1}^{r-1}$$



〖Example 10〗Determine the number of ways to insert tokens worth $1,$2 and $5 into a vending machine to pay for an item that costs *r* dollars in both the cases when the order in which the tokens are inserted does not matter and when the order does matter.



Solution:

(1) The order in which the tokens are inserted does not matter

$$G(x) = (1 + x + x^2 + x^3 + \cdots)(1 + x^2 + x^4 + x^6 + \cdots)(1 + x^5 + x^{10} + x^{15} + \cdots)$$

The coefficient of x^r in the expansion of $G(x)$ is the solution of this problem.

(2) The order in which the tokens are inserted does matter

- **The number of ways to insert exactly *n* tokens to produce a total of *r*$ is the coefficient of x^r in $(x + x^2 + x^5)^n$**
- **Since any number of tokens may be inserted, the number of ways to produce *r*$ using $1,$2 and $5 tokens, is the coefficient of x^r in**

$$1 + (x + x^2 + x^5) + (x + x^2 + x^5)^2 + \ldots = \frac{1}{1 - (x + x^2 + x^5)}$$



5. Using Generating Functions to Solve Recurrence Relations

The Methods of Solving Recurrence Relations

- **Iterative approach**
- **Use a systematic way to solve an important class of recurrence relations**
- **Generating functions**

Method:

(1) Use the recurrence relation to find the generating function of this sequence;

(2) $G(x) \leftrightarrow a_n$



〖Example 11〗 Use generating functions to solve the recurrence relation $a_n = 2a_{n-1} + 3a_{n-2} + 4^n + 6$ **with initial conditions** $a_0 = 20, a_1 = 60$.

Solution:

$$a_n = 2a_{n-1} + 3a_{n-2} + 4^n + 6 \qquad \times x^n$$

$$a_n x^n = 2a_{n-1}x^n + 3a_{n-2}x^n + 4^n x^n + 6x^n$$

$$\sum_{n=2}^{\infty} a_n x^n = 2\sum_{n=2}^{\infty} a_{n-1}x^n + 3\sum_{n=2}^{\infty} a_{n-2}x^n + \sum_{n=2}^{\infty} 4^n x^n + 6\sum_{n=2}^{\infty} x^n$$

$$G(x) - a_0 - a_1 x \qquad 2x\sum_{n=1}^{\infty} a_n x^n \qquad 3x^2\sum_{n=0}^{\infty} a_n x^n \qquad \frac{1}{1-4x} - 1 - 4x \qquad 6\left(\frac{1}{1-x} - 1 - x\right)$$

$$2x(G(x) - a_0) \qquad 3x^2 G(x)$$



$$(1-2x-3x^2)G(x) = \frac{20-80x+2x^2+40x^3}{(1-4x)(1-x)}$$

$$G(x) = \frac{20-80x+2x^2+40x^3}{(1-4x)(1-x)(1+x)(1-3x)}$$

$$= \frac{16/5}{1-4x} + \frac{-3/2}{1-x} + \frac{31/20}{1+x} + \frac{67/4}{1-3x}$$

$$\frac{16}{5}\times 4^n \quad -\frac{3}{2}\times 1^n \quad \frac{31}{20}\times(-1)^n \quad \frac{67}{4}\times 3^n$$

$$a_n = \frac{16}{5}\times 4^n - \frac{3}{2} + \frac{31}{20}\times(-1)^n + \frac{67}{4}\times 3^n$$



6. Proving Identities via Generating Functions

The method of proving combinatorial identities:

- **Use combinatorial proofs**
- **Use generating functions**



〖Example 12〗 Use generating functions to prove Pascal's identity $C(n,r) = C(n-1,r) + C(n-1,r-1)$ when *n* and *r* are positive integers with $r < n$.



Proof:

$$G(x) = (1+x)^n = \sum_{r=0}^{n} C(n,r)x^r$$

$$(1+x)^n = (1+x)(1+x)^{n-1} = (1+x)^{n-1} + x(1+x)^{n-1}$$

$$\sum_{r=0}^{n} C(n,r)x^r = \sum_{r=0}^{n-1} C(n-1,r)x^r + \sum_{r=0}^{n-1} C(n-1,r)x^{r+1}$$

$$= \sum_{r=0}^{n-1} C(n-1,r)x^r + \sum_{r=1}^{n} C(n-1,r-1)x^r$$

$$1 + \sum_{r=1}^{n-1} C(n,r)x^r + x^n$$

$$= 1 + \sum_{r=1}^{n-1} C(n-1,r)x^r + \sum_{r=1}^{n-1} C(n-1,r-1)x^r + x^n$$

$$\sum_{r=1}^{n-1} C(n,r)x^r = \sum_{r=1}^{n-1} [C(n-1,r) + C(n-1,r-1)]x^r$$



Homework: (Due on May 5th)

Ver. 8

Sec. 8.4 6(d,f), 10 (c, d, e), 16, 24(a), 32, 36, 45

Ver. 7

Sec. 8.4 6(d,f), 10 (c, d, e), 16, 24(a), 30, 34, 43

